OASP UE, WS 2026

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JOHANNES KEPLER UNIVERSITÄT LINZ

INSTITUTE OF SIGNAL PROCESSING

Course Information

General Information

Exercise Course	Optimum and Adaptive Signal Processing Systems		
Course head:	Dr. Matthias Wagner, DI Bernhard Plaimer		
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Course:	382.018	382.019	382.026
Room:	K 034D	HS 18, K 033C, HS 17, HS8	HS 5
Time:	Thu, 12:00 - 13:30	Thu, 13:45 - 15:15	Thu, 15:30 - 17:00

Exercise Courses

During the problem classes, practical examples in line with the lecture content will be presented. At every occurrence of the problem classes a new homework assignment will be presented and put onto the Moodle course page. These homework assignments have to be solved in **teams of two students** at home. These teams are fixed with the first homework assignment. Please clearly indicate your team on the cover of the first homework assignment. You will usually have two weeks time for each homework assignment.

The assignments cover both analytical and numerical exercises. The analytic exercises are marked with ✎, and the numerical exercises are marked with symbols.

Numerical Exercises

The numerical exercises can be carried out in MATLAB, OCTAVE, or in PYTHON. This gives you the freedom to choose your most convenient programming language to work out this course. This freedom comes with some responsibilities:

- Please ensure that every code file is executable! We cannot debug your code.
- Some numerical exercises contain code provided by us. This code is usually provided for MATLAB. It is your responsibility to find equivalent code for your preferred programming language.

Submission

The whole report and all scripts need to be uploaded to Moodle (one upload per team).

All files for submission must be zipped (.zip) and following name assigned:
OASP_UE_Assignment<i>_<MatNr1>_<MatNr2>.zip

The corresponding report containing all computations, plots, answered questions, etc. must be a **single** PDF document (**containing a title page**) within the zip file:
OASP_UE_Assignment<i>_<MatNr1>_<MatNr2>.pdf
Handwritten sheets can be scanned or photographed but must be merged into this document

in correct order.

Submission checklist
Before each submission, verify the following points:

- Ensure all scripts are executable. Add a proper description in the beginning of the file. For each problem there must be a **single** script file, e.g., `problem1.m` (which of course could call other scripts/functions).
- Do not put source code into your report
- Ensure all figures have proper labels, etc.
- Ensure the notation (see below) is met
- Please only use the % symbol for comments within an octave program code

Grading

The grading is based on the submitted homework assignments. The deadline for submission will be provided on each homework assignment.
Each homework assignment has a maximum of 40 points that can be achieved. There will be 6 homework assignments. Therewith the following grading applies:

Sehr Gut (1)	Gut (2)	Befriedigend (3)	Genügend (4)	Nicht Genügend (5)
$\geq$ 210 pts	$\geq$ 180 pts.	$\geq$ 150 pts.	$\geq$ 120 pts.	$<$ 120 pts.
$\geq$ 87.5 %	$\geq$ 75 %	$\geq$ 62.5 %	$\geq$ 50 %	$<$ 50 %

Table 1: Grading

Notation

- For signal representations, use (.) for continuous and [.] for discrete variables
- Vector representation: Digital: $\mathbf{x}$, Hand written: $\underline{x}$ [1]
- Matrix representation: Digital: $\mathbf{X}$, Hand written: $\underline{\underline{X}}$

Plagiarism

The homework assignments have to be solved individually from the teams. In case plagiarism is encountered there will be no points granted for all involved teams, no exclusions.

[1] Exception: Assignment 1/2, as random variables are denoted with upper case letters.